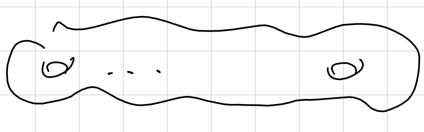
S^n, ~~$S^1 \times S^1$~~, $\mathbb{RP}^n$,



$(n \geq 2)$
$\downarrow$
$\pi_1 = 0$.

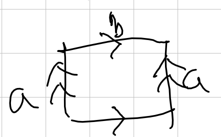
2-dim mfd

$L(p,q)$

$S^1 \times S^1 / e^{2\pi i p}, e^{2\pi i q}$

$M \# N$. ∧ ∨

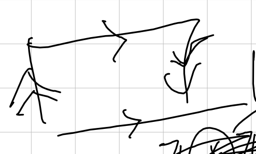
$[0,1]^2$



torus.

$\mathbb{Z} \times \mathbb{Z}$

Klein.

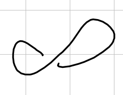


Klein. free product.

~~$\mathbb{Z} * \mathbb{Z}$~~

abababb..

$\langle a,b \mid aba = b \rangle$.



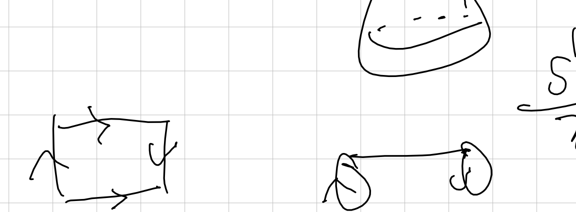
$aba^{-1}b^{-1} = e$.

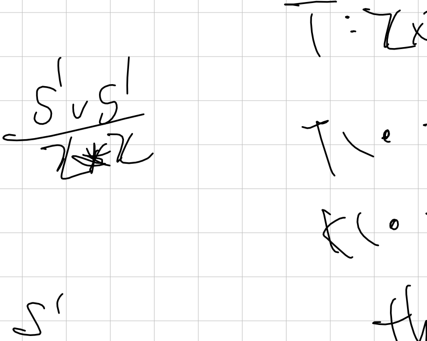
$\langle a,b \mid ab = ba \rangle$

$\mathbb{Z} \times \mathbb{Z}$

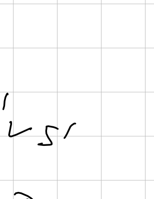
$ab = ba$ → grmp +ℤ.

$S^1 \vee S^1$

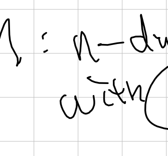




$S^1 \vee S^2$



$T : \mathbb{Z} \times \mathbb{Z}$



$S^1 \vee S^1$
$\mathbb{Z} * \mathbb{Z}$

$T \setminus \cdot$ $\mathbb{Z} * \mathbb{Z}$

$K \setminus \cdot$ $\mathbb{Z} * \mathbb{Z}$

theorem

Theorem

$S^1 \vee S^1$
$\mathbb{Z} * \mathbb{Z}$

S^1
$\updownarrow$
$\mathbb{Z}$

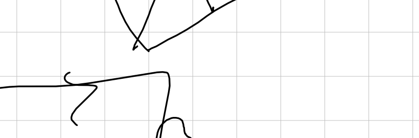
$S^3 : 0$.
$\downarrow$
$\mathbb{R}^3 : 0$.

M : n-dim
with $n \geq 3$

then $\pi_1(M) \simeq \pi_1(M \setminus \cdot)$.
$\simeq \pi_1(M \setminus 2pt)$
$\simeq \pi_1(M \setminus \{x_1, \ldots\})$

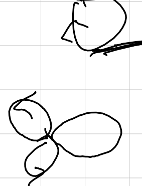
$\pi_1(S^n) = 0$
$n \geq 2$.

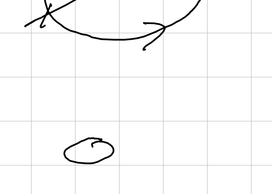
$\pi_1(\mathbb{RP}^n) = \mathbb{Z}_2$
$(n \geq 2)$.

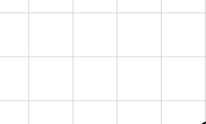


~~tree~~

$S^1 \vee S^1 \vee \mathbb{R}$.
$\downarrow$
$\mathbb{Z} * \mathbb{Z}$

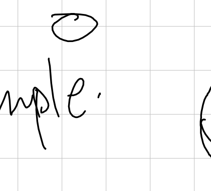


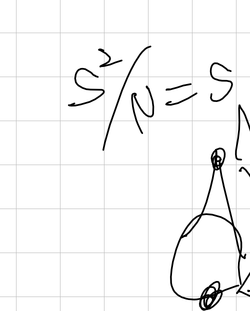


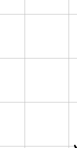


$: S^1 \vee S^1 \vee S^1$

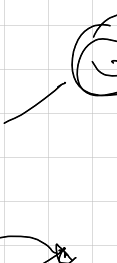
$\mathbb{Z} * \mathbb{Z} * \mathbb{Z}$



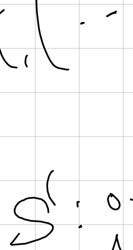


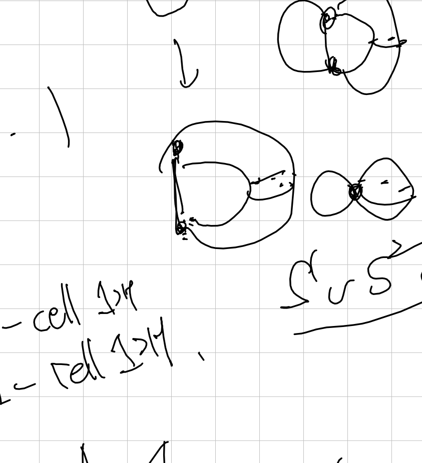


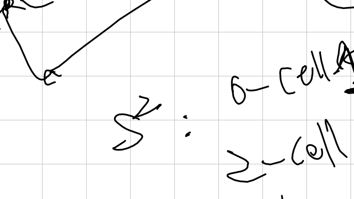
Example.



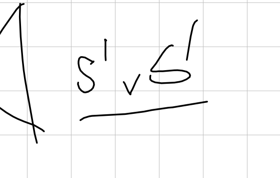
$S^2 / N = S$



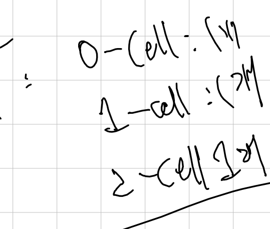


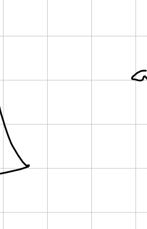


$\rightarrow \pi_1(\ldots)$



$S^1 \vee S^2$,





S^1 : 0-cell 1개
1-cell 1개.

S^2 : 0-cell 1개
2-cell 1개

$S^1 \vee S^1$

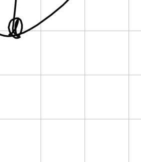
$\mathbb{RP}^2$: 0-cell : 1개
1-cell : 1개
2-cell 1개

$aba^{-1}b^{-1} = 0$.

$\langle a,b \mid ab = ba \rangle$

$\mathbb{Z} * \mathbb{Z} / \alpha$

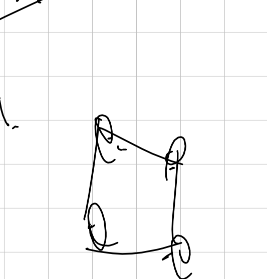
$\mathbb{RP}^2 : \mathbb{Z}_2$ $\langle a \mid a^2 \rangle$
$a^2 = 1$

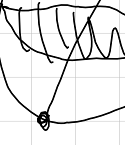


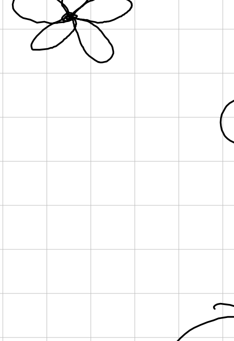
$\mathbb{RP}^3$: ~~0-cell 1개~~
~~1-cell 1개~~
~~2-cell 1개~~
~~3-cell 1개~~

~~$\pi_1(\ldots)$~~

$\pi_1(\mathbb{RP}^3) = \pi_1(\mathbb{RP}^2) = \mathbb{Z}_2$.

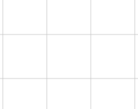


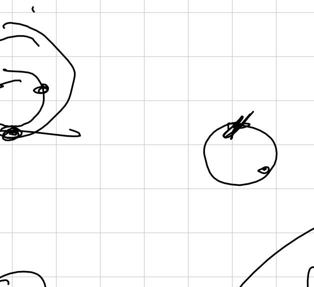


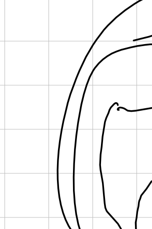


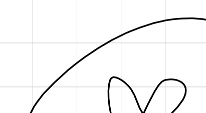
← $x^2+y^2+z^2=1$
∪ $x^2+y^2 \leq 1$
∪ $(0,0,z)$ $-1 \leq z \leq 1$

$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

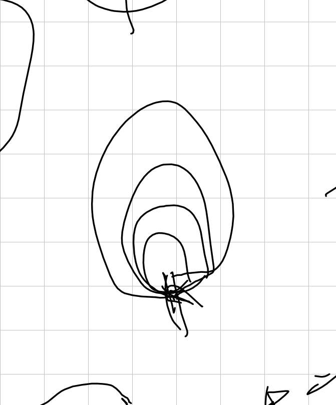


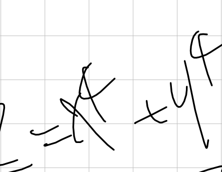


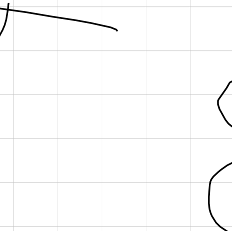




$\mathbb{R}/\mathbb{Z}$

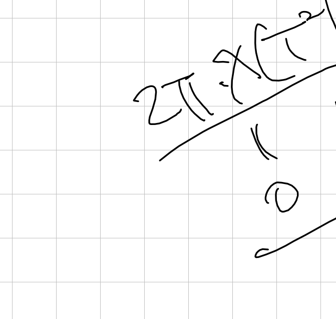






$z = x^4 + y^4$
↓ exp

$K_1 = f(x, 2)$
$K_2 = f_y$



$p = 0$

$\frac{1}{n+z}$.

$\int \frac{Rdz}{y}$ $\int \frac{1}{z} dz$
$2\pi i$

$2\pi i \times (-1)$
0